

Wobfuscator: Obfuscating JavaScript Malware via Opportunistic Translation to WebAssembly

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Content

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--- page 1 ---

Wobfuscator: Obfuscating JavaScript Malware via Opportunistic Translation to WebAssembly Alan Romano University at Buffalo, SUNY Buffalo, United States alanroma@buffalo.edu Daniel Lehmann University of Stuttgart Stuttgart, Germany mail@dlehmann.eu Michael Pradel University of Stuttgart Stuttgart, Germany michael@binaervarianz.de Weihang Wang University at Buffalo, SUNY Buffalo, United States weihangw@buffalo.edu Abstract —To protect web users from malicious JavaScript code, various malware detectors have been proposed, which analyze and classify code as malicious or benign. State-of-the-art detectors focus on JavaScript as the only target language. However, WebAssembly provides attackers a new and so far unexplored opportunity for evading malware detectors. This paper presents Wobfuscator, the first technique for evading static JavaScript malware detection by moving parts of the computation into WebAssembly. The core of the technique is a set of code transformations that translate carefully selected parts of behavior implemented in JavaScript into WebAssembly. The approach is opportunistic in the sense that it uses WebAssembly where it helps to evade malware detection without compromising the correctness of the code. Evaluating our approach with a dataset of 43,499 malicious and 149,677 benign JavaScript files, as well as six popular JavaScript libraries reveals that our approach is effective at evading state-of-the-art, learning-based static malware detectors; the obfuscation is semantic-preserving; and our approach has small overhead, making it practical for use in real-world programs. By pinpointing limitations of current malware detectors, our work motivates future efforts on detecting multi-language malware in the web. I. I

INTRODUCTION The omnipresence of the web makes client-side web applications an attractive target for attackers. As a result, various kinds of attacks target the browser, e.g., drive-by malware [48], [23], [43], malicious code deployed via script-based browser augmentation markets [58],

browser-based cryptomining without user consent [40], [52], [34], malicious browser extensions [21], [65], and browser-based phishing [14]. A recent report estimates that orchestrated phishing campaigns alone create 1.7 to 2 million malicious payload URLs each month [44]. Beyond such obviously malicious activities, many websites employ techniques that are unwanted by users, such as extensive browser fingerprinting [30] or tracking [25]. To protect users against executing malicious scripts in their browser, JavaScript malware detectors warn about such scripts before or while executing them. One line of work statically analyzes scripts before they are executed [24], [50], [56], [28], [27], e.g., by intercepting them already in the network, as part of a browser, or as part of a separate anti-virus tool. Another line of work dynamically analyzes scripts [36], e.g., by instrumenting the code or via a browser extension. To reduce the runtime overhead imposed by dynamic analysis-based malware detectors, static detectors often serve as a first line of defense, e.g., by dynamically analyzing only those scripts that are deemed dangerous by a static analysis. To effectively attack users despite the presence of malware detectors, attackers try to hide the maliciousness of scripts via obfuscation and evasion techniques [53], [55]. Progress by attackers and defenders leads to an arms race between increasingly sophisticated obfuscation and evasion techniques on one hand and increasingly effective malware detectors on the other hand. Currently, the most effective malware detection techniques use learning-based classifiers to distinguish malicious from benign scripts [24], [50], [28], [27]. These approaches extract a set of features from a given JavaScript file, e.g., n-grams of code tokens or AST-based features, or feed the JavaScript code into a deep neural network [63] to determine whether the file is likely to be malicious. While the focus on JavaScript historically makes sense, JavaScript is not the only language of the client-side web anymore. WebAssembly [31] is another language that is widely available in browsers. First announced in 2015, WebAssembly is supported by all major browsers since November 2017, and available in 94% of all global browser installations as of August 2021 [1]. The language provides an efficient compilation target for computation-intensive libraries written in languages such as C and C++. In addition to the many positive uses of WebAssembly, the language provides a new opportunity to attackers for evading malware detectors – an opportunity that, to the best of our knowledge, has not yet been explored. This paper presents the first technique for evading JavaScript malware detection by moving parts of the computation into WebAssembly. We describe Wobfuscator, a code obfuscation technique that transforms a given JavaScript file into a new JavaScript file and a set of WebAssembly modules. By changing the malicious JavaScript code, our work aims at evading static malware detectors. The rationale is that static detectors may be used on their own or serve as a filter for which scripts to analyze dynamically. That is, evading static malware detectors gives a huge benefit to attackers. Transforming parts of a JavaScript file into WebAssembly is far from trivial. JavaScript is dynamically typed, has complex objects, and provides direct access to browser APIs. In contrast, WebAssembly is statically typed, has only four low-

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level, primitive data types, and it can access browser APIs only indirectly by importing them from JavaScript. Because of these fundamental differences, general JavaScript-to-WebAssembly translation is practically impossible, which is reected in the fact that WebAssembly never has been touted as a replacement for JavaScript but as a way to complement it [31]. The core technical contribution of this paper is a set of code transformations that extract carefully selected parts of behavior implemented in JavaScript for translation into WebAssembly. The approach is opportunistic in the sense that it translates JavaScript to WebAssembly where it helps to evade malware detectors, without compromising the correctness of the code. For example, we present a transformation that extracts function calls into a WebAssembly module, obfuscating if and when a script calls a particular function. Other transformations aim at obfuscating string literals, control-flow statements, and array initializations. Preconditions guard each transformation to ensure the original behavior is preserved, a property we consider crucial for an attacker to use an obfuscation technique. Our work relates to existing obfuscation techniques for JavaScript. For example, Fass et al. fuse malicious code into benign code while preserving the AST of the benign code [26]. For a more comprehensive overview of obfuscation techniques and their prevalence, we refer readers to other work [66], [55]. All of these approaches obfuscate code via transformations within the JavaScript language, whereas Wobfuscator exploits the availability of WebAssembly to obfuscate code by translating beyond JavaScript. A paper by Wang et al. shares the general idea of obfuscating code written in one language by translating parts of it into another language [60]. However, they describe partially translating C code into Prolog, which does not address the unique challenges of obfuscating JavaScript via partial translation to WebAssembly. We evaluate Wobfuscator with a dataset of 43,499 malicious and 149,677 benign JavaScript files, as well as six popular JavaScript libraries. Our results show the following. First, the approach is effective at evading state-of-the-art, learning-based static malware detectors: Applying our transformations reduces the recall of the four studied detectors [50], [24], [28], [27] to 0.18, 0.63, 0.18, and 0.00, respectively. Second, the obfuscation preserves the semantics of the transformed code: Obfuscating six popular JavaScript libraries and running their 2,017 tests shows no observable changes in the behavior of the tested code. Finally, we find that our tool only takes on average 8.9 seconds to apply all the transformations to a project (with on average 4,152 lines of code) and adds on average 31.07% of overhead during runtime. Overall, these results show that Wobfuscator is practical for use in real-world programs. In summary, this paper contributes the following:

The first technique to use WebAssembly as a means for obfuscating the behavior of malicious JavaScript code.

A set of code transformations that translates carefully selected JavaScript code locations into WebAssembly.

A comprehensive evaluation showing that the approach effectively evades state-of-the-art static malware detectors while preserving the semantics of the original code. Our experiment results are publicly available: <https://github.com/js2wasm-obfuscator/translator>. II. BACKGROUND ON WEB ASSEMBLY We present a brief introduction of the core concepts and syntax of WebAssembly. For space reasons, we refer the reader to the original publication [31], official website [10], or

specification [64] for more elaborate explanations. WebAssembly is a low-level byte code language. WebAssembly programs are distributed as binaries that are compact to send over the network and quick to parse. Each WebAssembly program is a single file module, organized into several sections. Most importantly, the code section contains all functions with their bodies. The global section contains scalar global variables. The memory section declares a linear, byte-addressable memory, of which parts can be initialized with the data section. WebAssembly instructions operate at a low level of abstraction, without, e.g., classes or complex objects. Instructions and functions are statically typed, but there are only four primitive types: 32- and 64-bit integer and floating-point values, respectively. Instructions are executed on a virtual stack machine, i.e., they pop their operands and push their results onto an implicit evaluation stack. Instructions are simple and designed to map closely to hardware instructions, e.g., a WebAssembly `i32.add` instruction would be translated into an x86 `addl`. Since there is no garbage collector and only primitive scalar types, complex objects (strings, arrays, records, etc.) are stored into program-organized linear memory, which is essentially a growable array of untyped bytes. WebAssembly does not have a standard library, and for interaction with the "outside world", WebAssembly modules need to import functions from the host environment. In the browser, that host environment is JavaScript, so any JavaScript function can be called from WebAssembly. Non-primitive data needs to be marshalled through the WebAssembly memory however (which can be accessed from JavaScript as well). The WebAssembly API in JavaScript provides functions to compile and instantiate (that is, fulfill the imports of) WebAssembly modules and call exported WebAssembly functions.

III. THREAT MODEL

Our work is about malicious or unwanted code delivered as part of a client-side web application. A website may deliver such code intentionally, e.g., because the domain is controlled by the attacker, or unintentionally, e.g., via third-party scripts or advertisements. We assume that the attacker controls the malicious code on the server side, and hence, can freely modify it. In particular, they can transform the source code, split one file into multiple files, or merge multiple code files. On the defense side, we assume a static malware detector that scans the client-side code of a web application before it executes in the browser. The mechanism the malware detector uses to intercept and check the code is irrelevant to us. For example, it may be implemented as a network proxy that scans

--- page 4 ---

```

Input: MaliciousJavaScriptOutput: JavaScript + WebAssembly
var foo = "evil"; eval(...); JavaScript
ASTApproachIdentify & checkTranslation sitesGenerate WebAssemblyRewrite JavaScript
then-US(module(memory...))
(func...)+InstantiationSiteAsyncWrapperthen-USTranslationSite

```

--- page 5 ---

Fig. 1. Overview of Wobfuscator. files before they even reach the client's machine, as an anti-virus tool on the client machine, or as a browser extension. Malware detection or mitigation techniques that go beyond static analysis, e.g., based on analyzing the executing of client-side code, are beyond the scope of this work. The goal of the attacker is to bypass the malware detector. To this end, the attacker may assume that the targeted browser supports WebAssembly, which is the case for almost all of today's browsers. The attacker does not need to have access to the malware detector or, in the case a machine learning-based detector, to the data it is trained on.

IV. APPROACH

A. Overview and Challenges

Figure 1 gives an overview of Wobfuscator. The input is a

JavaScript file, which we parse into an AST. Next, the approach identifies potential translation sites, i.e., code locations that (i) are relevant for detecting malicious code and (ii) can be translated into WebAssembly in a semantics-preserving way. Instead of aiming at a general JavaScript-to-WebAssembly translation, the approach opportunistically targets only those code locations that fulfill these two requirements. To move behavior into WebAssembly, Wobfuscator generates WebAssembly code for each translation site and then transforms the JavaScript AST to utilize the generated code. The AST is transformed in three ways. First, at an instantiation site, we add code to load the WebAssembly module into the application. Second, at each of the selected translation sites, we modify the code to access properties and functions provided by the WebAssembly module(s). Third, at the root node of the AST, we conditionally wrap the script into an anonymous, async function, referred to as the async wrapper, to support asynchronous keywords in the code. The remainder of this section explains these transformations in detail. Finally, the output of Wobfuscator is the transformed JavaScript code along with one or more generated WebAssembly modules. We encountered several challenges in designing transformations that move JavaScript behavior into WebAssembly modules. One key challenge is the fact that completely translating general JavaScript code to WebAssembly is impossible due to the limited set of features in WebAssembly. One example is dynamically generated code, which is enabled in JavaScript through the infamous `eval` function, but not supported in WebAssembly. JavaScript supports rather complex rules, e.g., function scope for `var`-bound variables, hoisting of functions, and closures. WebAssembly has only three storage locations: global module variables, local variables for each function, and raw byte memory. In addition, WebAssembly only has limited control-flow instructions that do not fully replicate all available JavaScript constructs, such as `try / catch` statements, `for-of` loops, and `Promises`. Another challenge is the limited data types that can be passed between WebAssembly and JavaScript. In the initial version of WebAssembly, functions can only accept and return numeric data types, which cannot replicate the complex objects that JavaScript supports. Finally, different browsers can impose file-size limits on the WebAssembly modules used, so we must work around this limitation for our technique to be general. Because of these and other differences, Wobfuscator is based not on complete but opportunistic translation, i.e., transforming code where it helps to evade malware detectors, without sacrificing correctness.

B. Transformations The goal of our approach is to generate WebAssembly modules that can reproduce the functional behavior of specific JavaScript code snippets. To produce these modules, Wobfuscator is constructed on a core set of transformation rules: Definition 1 (Transformation rule). A transformation rule is a tuple $(L; t; p)$ where:

L represents a set of code locations where the transformation may apply,

t is a transformation function that maps JavaScript at a code location in L to rewritten JavaScript code and one or more WebAssembly modules, and

p is a precondition for applying t , expressed as a predicate on a code location and its surrounding context. We present seven transformation rules that target different language features of JavaScript. The transformation rules fall into three categories. First, we present rules to obfuscate data that gets defined and used by a JavaScript file. These rules target string literals (Section IV-B1) and array initialization code (Section IV-B2). Second, we present rules to obfuscate function calls. These rules hide suspicious function calls (Section IV-B3) or any function call (Section IV-B4). Finally, we present rules to obfuscate the control flow in the given code. These rules target if

statements (Section IV-B5), for loops (Section IV-B6), and while loops (Section IV-B7). Table I illustrates the transformation rules. The transformation rules use several JavaScript primitives to interact with WebAssembly:

`instanWasm(source, impObj)` instantiates a Web-Assembly module from `source` and returns the module. The optional parameter `impObj` is an object containing the functions to be imported into the created WebAssembly module. We use two variants of this primitive which instantiate the module synchronously and asynchronously (Section IV-C).

`loadStrFromBuf(buffer, startIndex)` creates a string from the buffer starting from the byte offset `startIndex` and ending at the first null byte (i.e., `n 00`)

--- page 6 ---

TABLE I TRANSFORMATION FUNCTIONS .Rule JavaScript (Before) JavaScript (After) WebAssembly (After)

T1-StringLiteral 1 var s = "lit"; 1 // Instantiation Site 2 let m = instanWasm(source); 3 let buf = m.instance.exports.memory.buffer; 4 // Translation Site 5 let startInd = m.instance.exports.d1; 6 var s = loadStrFromBuf(buf, startInd); 1 (global \$d1 (export "d1") (mut i32) (i32.const 0)) , ! 2 (memory (export "memory") 1) 3 (data \$data0 (i32.const 0) "lit\00"))

T2-ArrayInitialization 1 var arr = new Array(); 2 arr[i1] = num1; 3 arr[i2] = num2; 1 // Instantiation Site 2 let m = instanWasm(source); 3 let buf = m.instance.exports.memory.buffer; 4 // Translation Site 5 m.instance.exports.f(); 6 var arr = loadArrFromBuf(buf, startInd, len); 1 (memory (export "memory") 1) 2 (func \$f (export "f") 3 i32.const \$i1 4 i32.const \$num1 5 i32.store 6 i32.const \$i2 7 i32.const \$num2 8 i32.store)

T3-FunctionName 1 eval(str); 1 // Instantiation Site 2 let m = instanWasm(source); 3 let buf = m.instance.exports.memory.buffer; 4 // Translation Site 5 let startInd = m.instance.exports.d1; 6 windowloadStrFromBuf(buf, startInd); 1 (global \$d1 (export "d1") (mut i32) (i32.const 0)) , ! 2 (memory (export "memory") 1) 3 (data \$data0 (i32.const 0) "eval\00")) , !

T4-CallExpression(a) 1 f(a); 1 // Translation Site 2 let impObj = {imports: {impFunc: () => f(a)}}; 3 let m = instanWasm(source, impObj); 4 m.instance.exports.f0(); 1 (func \$f0 (export "f0") 2 call \$impFunc) ;; JS import

T4-CallExpression(b) 1 let r = f(a); 1 // Translation Site 2 let impObj = {imports: {impFunc: f}}; 3 let m = instanWasm(source, impObj); 4 let r = m.instance.exports.f0(a); 1 (func \$f0 (export "f0") (param externref) (result externref) , ! 2 local.get \$p 3 call \$impFunc) ;; JS import

T5-IfStatement 1 if(cond) { 2 stmt1; stmt2; ... 3 } else { 4 stmt3; stmt4; ... 5 } 1 // Translation Site 2 let impObj = {imports: { 3 imp1:() => {stmt1; stmt2; ...}, 4 imp2:() => {stmt3; stmt4; ...}}}; 5 let m = instanWasm(source, impObj); 6 m.instance.exports.f(cond ? 1 : 0); 1 (func \$f (export "f0") (param \$p) 2 local.get \$p 3 if ;; label = @1 4 call \$imp1 ;; JS import 5 else 6 call \$imp2 ;; JS import 7 end)

T6-ForStatement 1 for(init;cond;incre) { 2 stmt1; stmt2; ... 3 } 1 // Translation Site 2 init; 3 let impObj = { 4 imports: { 5 cond:() => {return cond ? 1 : 0}, 6 incre:() => {incre}, 7 body:() => {stmt1; stmt2; ...} 8 } 9 }; 10 let m = instanWasm(source, impObj); 11 m.instance.exports.f(); 1 (func \$f (export "f0") 2 block \$L0 3 loop \$L1 4 call \$cond ;; JS import 5 i32.eqz 6 br_if \$L0 7 call \$body ;; JS import 8 call \$incre ;; JS import 9 br \$L1 10 end 11 end)

T7-WhileStatement 1 while(cond) { 2 stmt1; stmt2; ... 3 } 1 // Translation Site 2 let impObj = { 3 imports: { 4 cond:() => {return cond ? 1 : 0}, 5 body:() => {stmt1; stmt2; ...} 6 } 7 }; 8 let m = instanWasm(source, impObj); 9 m.instance.exports.f(); 1 (func \$f (export "f0") (param \$p) 2 block \$L0 3 loop \$L1 4 call \$cond ;; JS import 5 i32.eqz 6 br_if \$L0 7 call \$body ;; JS import 8 br \$L1 9 end 10 end)

after `startIndex` , where `buffer` is the WebAssembly module linear memory.

`loadArrFromBuf(buffer, startIndex, length)` creates an array from buffer of size `length` starting from the byte offset `startIndex` , where `buffer` is the WebAssembly linear memory.

1. Obfuscating String Literals:

JavaScript malware frequently uses encoded strings to hide malicious code [66]. These encoded strings can be critical for malware detectors which learn the string patterns and their encoding schemes [24], [54]. To evade the detection of encoded strings, we define a transformation rule T1-StringLiteral ($L \ T \ 1$; $t \ T \ 1$; $p \ T \ 1$) where: $L \ T \ 1$. The code locations where transformation rule T1 may apply are all AST nodes of Literal type with string values. $t \ T \ 1$. The transformation function is defined in row T1-StringLiteral in Table I. To obfuscate a string literal "lit", $t \ T \ 1$ generates a WebAssembly module that defines a memory and exports it to JavaScript (line 2). The memory is used to store the string literal "lit" at offset 0 (line 3). To reconstruct this string in JavaScript, a variable \$d1 containing the offset is defined and exported (line 1). Each string is terminated with a null byte (i.e., `\0`), so it can be reconstructed by reading the linear memory from the starting index until the first null byte is found. If multiple strings in the input JavaScript program are to be transformed, they are all stored in a single WebAssembly module. In this case, multiple variables can be defined for each string stored (e.g., \$d1, \$d2, ...). The variable buf points to the 4

--- page 7 ---

memory exported by WebAssembly (line 3). At the translation site, a variable startInd gets the starting index of the string literal stored in linear memory (line 5). Finally, the string "lit" is reconstructed using the primitive loadStrFromBuf with arguments buf and startInd (line 6). $p \ T \ 1$. This transformation can be applied in locations where JavaScript allows for replacing a string literal with a function call. Specifically, this excludes: (i) string literals used in import or require statements, as such strings should be known when bundling modules together; (ii) string literals used as property names in object expressions as return values from function calls cannot be used as object keys.

1. Obfuscating Arrays:

Malicious files often contain arrays of numeric literals representing character codes used to reconstruct malicious strings. To obfuscate these arrays, we exploit the fact that the linear memory of WebAssembly modules is implemented through an array buffer, which can naturally map to a JavaScript numeric array, in transformation rule T2-ArrayInitialization ($L \ T \ 2$; $t \ T \ 2$; $p \ T \ 2$) where: $L \ T \ 2$. T2 may apply on NewExpression AST nodes (e.g., `new Array`) or ArrayExpression AST nodes representing array literal expressions (e.g., `[1,2,3]`). In addition, the transformation also condenses any following AssignmentExpression nodes used to initialize the array values into a single JavaScript function call (e.g., `arr[1] = 42; arr[2] = 97; ...`). $t \ T \ 2$. The transformation $t \ T \ 2$ is defined in row T2-ArrayInitialization in Table I. The original JavaScript code creates an array arr and initializes the elements arr[i1] and arr[i2] with numerical values num1 and num2, respectively. After the transformation, $t \ T \ 2$ produces a WebAssembly module that creates a memory and exports it (line 1). A function \$f is defined which stores the numbers, num1 and num2, at the specified offsets, i1 and i2, inside the linear memory (lines 2-8). The transformed JavaScript code instantiates a WebAssembly module (line 2) and creates a variable to read from the exported memory (line 3). Similar to T1-StringLiterals, if there are multiple array initializations to be transformed, only one WebAssembly module is created at the instantiation site. At the translation site, the export function f is called to write num1 and num2 to the linear memory at offset i1 and i2 (line 5). Finally, loadArrFromBuf() returns a JavaScript Array object containing the numerical values copied from the linear memory buffer, and this array is assigned to arr (line 6). This function

requires the starting index of the array in linear memory and the length of the array. Both of these values are calculated by Wobfuscator and inserted into each call to `loadArrFromBuf()`. We ensure that a JavaScript Array object is returned as the original code using the array will need to access the standard properties and methods of an Array [2]. This transformation can be applied to arrays initialized with numeric literals. Since WebAssembly only has numeric data types we only store numeric literals in memory using the operators `.const` and `.store`. Specifically, we apply the transformation only if the array initialization is one of the following: (i) a new Array expression followed by assignment statements inserting only numeric literals; (ii) a new Array expression with numeric literal arguments; (iii) an ArrayExpression only containing numeric literals.

1. Obfuscating Function Names:

Several built-in JavaScript functions, such as the notorious `eval` function, are commonly exploited by attackers. As a result, detectors may consider the names of these built-in functions suspicious and use them as part of the signatures for malware detection [50], [24], [17]. To evade detection, we remove suspicious function names from the JavaScript code through a transformation rule T3-FunctionName (L T 3 ; p T 3 ; t T 3) where: L T 3 . T3 may apply on CallExpression nodes or NewExpression nodes that contain specific identifier names. t T 3 . The transformation function is defined in row T3-FunctionName in Table I. To obfuscate the function name `eval`, t T 3 removes the function name from JavaScript and stores it in WebAssembly linear memory. In the WebAssembly code, a global variable `$d1` is defined and exported with a value of 0 (line 1), which is the starting index of the function name "eval" stored in the linear memory. Next, a memory is created and exported (line 2). To initialize the linear memory, a data section is defined that contains "eval" at offset 0 (line 3). In the transformed JavaScript code, it instantiates a WebAssembly module (line 2) and defines a variable to access the linear memory (line 3). The variable `startIdx` is assigned the value of the exported `d1` that contains the starting index of "eval" in the linear memory (line 5). Finally, `loadStrFromBuf` is used to create the string "eval", and `eval` is called from the window object with `str`, the same argument used in the original `eval()` (line 6). p T 3 . This transformation can be applied to call expressions or new expressions referencing global functions accessible through the window object. Specifically, we identify eight global functions commonly used in malicious files and apply the transformation only if the identifier is in the following list: `eval`, `escape`, `atob`, `btoa`, `WScript`, `unescape`, `Function`, and `ActiveXObject`. While these functions are not inherently malicious, many of the analyzed malware files use these functions to decode and execute hidden code.

1. Obfuscating Calls:

Aside from the suspicious functions, we construct a transformation rule T4-CallExpression for general JavaScript function calls. This transformation converts function calls in JavaScript into a call of an exported WebAssembly function, which in turn performs a function call in WebAssembly to an imported JavaScript function. Unlike T3-FunctionName, which completely removes suspicious function names from JavaScript, this transformation modifies the context of the function being used in call sites that AST-based malware detectors use when scanning for malicious code. There is a trade-off in this transformation between compatibility with the WebAssembly Minimum Viable Product (MVP) version and the amount of transformable function calls. Thus, we create two variations of this transformation: T4-CallExpression(a) is fully compatible with the WebAssembly

MVP (i.e., uses no language extensions) but can only be applied on functions that do not return a value; T4- CallExpression(b) transforms functions with return values but 5

--- page 8 ---

requires the WebAssembly Reference Types proposal [9]. Firefox and Chrome enable this proposal by default. T4-CallExpression(a) is a transformation rule (L T 4 a ; p T 4 a ; t T 4 a) where: L T 4 a . L T 4 a are CallExpression nodes containing the identifier and arguments of a function call. t T 4 a . t T 4 a is defined in row T4-CallExpression(a) in Table I. To obfuscate a JavaScript function call f(a) , t T 4 a moves the function call into an anonymous function that is imported into WebAssembly (line 2). At the original call site, a WebAssembly export function f0 is called (line 4). In the WebAssembly code, the function that wraps the JavaScript function call is imported as \$impFunc . The function \$f0 is exported and calls the imported function \$impFunc (lines 1-2). Note that the anonymous function used to wrap the original JavaScript function always has the same type signature, i.e., a void function with no parameters. Thus, the same WebAssembly module can be compiled once and reused for every replaced function call, changing only the import object containing the appropriate JavaScript function. p T 4 a . This transformation can be applied to locations where the call expressions do not have a return value that is used in an assignment or in another expression, as the primitive data types of WebAssembly, i32 , i64 , f32 , f64 cannot represent all possible JavaScript function return values. The generalized variant T4-CallExpression(b) is a transformation rule (L T 4 b ; p T 4 b ; t T 4 b) where: L T 4 b . L T 4 b is the same as L T 4 a . t T 4 b . t T 4 b is defined in row T4-CallExpression(b) in Table I. The experimental WebAssembly Reference Types proposal [9] adds a new WebAssembly value type, externref , that can be used to pass references of arbitrary JavaScript values to and from WebAssembly. With this new type, the transformation only needs to import a reference to the transformed function. Specifically, the transformed JavaScript code only imports the function reference f into WebAssembly (line 2). In the original call expression, the function f is replaced with a WebAssembly export function f0 that takes in the argument a of the original call and returns any value that the original function outputs (line 4). In the WebAssembly code, the export function \$f0 takes in a parameter of type externref and returns a value of type externref (line 1). Inside \$f0 , it calls the imported function \$impFunc with a value that is the argument passed into \$f0 (lines 2-3). The benefits of this transformation over T 4 a include supporting more function calls by including those with return values and moving more behavior into WebAssembly than T 4 a . p T 4 b . This variant of the call transformation can be applied only if: (i) the callee function does not contain a reference to this ; (ii) the arguments of the call expression cannot contain a reference to this ; (iii) the data dependencies of the variables referenced within the callee cannot contain this ; (iv) the callee function is not a method of a literal value; and (v) the callee function is not the special functions bind or super . Conditions (i)-(iv) must be met because the value of this is changed when calling the function from within the WebAssembly module, which can lead to incorrect behavior. Condition (v) is needed as bind and super have restrictions on how they are called, so these functions cannot be passed as imports to the WebAssembly module.

1. Obfuscating If Statements:

By leveraging the control-flow instructions in WebAssembly, the behavior of if-else statements in JavaScript can be moved to WebAssembly, removing the syntactic information while preserving the semantics. To this end, we present transformation rule T5- IfStatement (L T 5 ; p T 5 ; t T 5)

where: $L \ T \ 5$. The transformation rule applies to `IfStatement` nodes. $t \ T \ 5$. The transformation is dened in row T5-`IfStatement` in Table I. At the translation site, $t \ T \ 5$ wraps the code within the if

- and

else -blocks in two anonymous functions that are imported into the WebAssembly module (lines 2-4). A WebAssembly export function f is called and the result of the test condition of the original if -statement is converted to a (zero or one) integer that is passed as the argument to f (line 6). Within the WebAssembly module, the two functions wrap- ping the code within the if

- and

else -blocks are imported as $\$imp1$ and $\$imp2$ (lines 4,6). The exported function $\$f$ takes in an integer parameter which will be either zero or one to act as a Boolean (lines 1-7). $\$f$ contains if-else instructions that are decided by the function parameter p . If p is non-zero, the if instruction calls $\$imp1$ that contains the code originally in the if -block. Similarly, if p is zero, then the else instruction calls $\$imp2$ containing the code originally in the else -block. By leveraging the if-else instructions in WebAssembly, the original semantics are preserved while the use of the control- ow statement is hidden from the JavaScript syntax. $p \ T \ 5$. The if statements can be transformed if the code blocks do not include the keywords `break` , `continue` , `return` , `yield` , or `throw` . These keywords are not compati- ble with moving the inner code blocks into functions.

1. Obfuscating For Loops:

To obfuscate for loops, we dene transformation rule $T6\text{-}ForStatement (L \ T \ 6 ; p \ T \ 6 ; t \ T \ 6) : L \ T \ 6$. The transformation locations where this rule applies are `ForStatement` nodes, which represent C-style for loops. $t \ T \ 6$. The transformation is dened in row T6-`ForStatement` in Table I. At the translation site, $t \ T \ 6$ wraps the loop condition, increment and body in three JavaScript functions that will be imported to the WebAssembly module (lines 3-8). The loop counter initialization can be hoisted out of the loop scope safely (line 2). Finally, a WebAssembly export function f is called which emulates the JavaScript for loop (line 11). In the WebAssembly module, the three JavaScript functions that are used to wrap the loop condition, increment, and body are imported as $\$cond$, $\$body$, and $\$incre$, respectively (lines 4, 7, 8). The export function $\$f$ contains a block of code with label $\$L0$, which encloses a loop block with label $\$L1$ (lines 2-3). Inside the loop , $\$cond$ is called (line 4) which evaluates the test condition of the JavaScript for loop and the result is checked using `i32.eqz` (line 5). If the test condition is false, the instruction `br_if $L0` branches out to the end of block $\$L0$, terminating the loop (line 6). Otherwise, if the test condition is true, functions $\$body$ and $\$incre$ are called 6

--- page 9 ---

to execute the statements within the loop body and the update expression (lines 7-8). The `br $L1` is used to branch to label $\$L1$, which continues iteration (line 9). $p \ T \ 6$. The precondition is the same as $p \ T \ 5$.

1. Obfuscating While Loops:

Analogous to the above, we obfuscate while loops with a transformation rule $T7\text{-}WhileStatement (L \ T \ 7 ; p \ T \ 7 ; t \ T \ 7)$ where: $L \ T \ 7$. The rule applies to `WhileStatement` nodes. $t \ T \ 7$. The transformation, dened in row T7-`WhileStatement` in Table I, is similar to $t \ T \ 6$. The only differences are that in $t \ T \ 7$, there is no loop increment nor loop counter initialization. A while loop can be

created by only wrapping the loop condition and the loop body into anonymous functions and importing them to WebAssembly (lines 4, 7). The function \$f is dened and exported, which emulates the JavaScript while loop by calling the import functions within the loop block. p T 7 . The precondition is the same as p T 5 and p T 6 . C. Synchronous and Asynchronous WebAssembly Instantiation For each transformation rule in Section IV-B (aside from T4-CallExpression(b)), we develop two variants differing in whether they instantiate the WebAssembly module synchronously or asynchronously, i.e., the implementation of the instanWasm() primitive. 1 let m = new WebAssembly.Module(2 new Uint8Array(decodeBase64(' ... '))); 3 return new WebAssembly.Instance(m, impObj); Fig. 2. Synchronous WebAssembly instantiation. The synchronous variants implement the primitive by using the WebAssembly.Module and WebAssembly.Instance constructor functions. Figure 2 shows the code, where in line 1, variable m is set to the compiled WebAssembly.Module object. The WebAssembly.Module constructor accepts a typed array containing the module bytes. Hence, we encode the module bytes into a base64 string, which is decoded to a typed Uint8Array at runtime (line 2). On line 3, the module object m , along with the import object, is then passed into the WebAssembly.Instance constructor. The returned instance object can be utilized by the transformation functions. Since this is standard, synchronous code, there are no restrictions on how to integrate instantiation into the original JavaScript application. However, because the WebAssembly.Module constructor can block the JavaScript main thread, browser vendors discourage this method. Chromium in particular even limits the input module to at most 4KB in size [6] and throws an exception otherwise. To get around this limitation, each synchronous transformation can emit one or more WebAssembly modules. Specically, transformations T3-FunctionName, T4- CallExpression(a), T5- IfStatement, T6-ForStatement, and T7- WhileStatement only emit one WebAssembly module per transformed. T1-StringLiteral, T2-ArrayInitialization, and T4- CallExpression(b) can emit one or more modules since the data stored within the modules can grow larger than 4KB. When a single WebAssembly module grows too large, e.g., because it contains many string literals in its data section, we split it into multiple modules to keep each under 4KB. For string literals larger than 4KB, the string literal is split across multiple modules and joined in the string reconstruction phase. 1 async function someFunction(){ 2 // Transformation site: ... 3 await (async () => { 4 let mod = await WebAssembly.instantiateStreaming(5 fetch("generated_module.wasm"), impObj); 6 //Transformation function code... 7 })() 8 } Fig. 3. Instantiation of the asynchronous variant. We also support asynchronous instantiation of the WebAssembly modules, which is the method that browser providers recommend. Asynchronous instantiation has several benets, including unrestricted module size and the ability to put the generated WebAssembly modules in separate les. These benets allow each transformation to only emit a single WebAssembly module. For these variants, the instanWasm() primitive is implemented via the WebAssembly.instantiateStreaming() function, as shown in Figure 3 (line 4). This API spawns compilation on a separate thread, thus not blocking the main thread of execution. Since this API returns a Promise , we need to add the await keyword to allow the Promise to resolve before continuing. Line 6 represents a placeholder for the transformation code of T1-T7. The await keyword can only be employed in asynchronous functions, so we wrap the instantiation in an async anonymous function (line 3). Similarly, since the anonymous function is an async function, its invocation must also have the await keyword added (line 3). The enclosing function, someFunction , now contains the await keyword, so the function denition must have the async keyword added (line 1). Elsewhere in the code, any function

calls to someFunction would also require adding the await keyword. As this example shows, inserting the async / await keywords to a translation site causes these keywords to be propagated to functions and call sites throughout the le. This keyword propagation makes the asynchronous transformations non-trivial to design and implement. Specifically, we encountered three code locations that are difficult to propagate the async / await keywords to. First, anonymous functions used as parameters in other function calls, e.g., .map , are difficult to handle. Depending on the return type of the anonymous function, the await keyword may need to be added within the called function's definition or to the function invocation. Second, class constructors cannot be made async , so a check must be done to detect if a constructor is in the call chain of any function. Third, if a transformed function is exported from a module, any other les using the function as an import must be checked for functions and function calls to add async and await to. All of the transformation rules have both synchronous and asynchronous variants except for T4-CallExpression(b). T4-CallExpression(b) relies on an experimental WebAssembly proposal that imposes complex preconditions. Adding 7

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the asynchronous restrictions to this may break the original semantics and lead to incorrect transformations. T4-CallExpression(b) exposes more translation sites than T4-CallExpression(a), increasing the number of edge cases that can be encountered. We leave this combination as future work. D. Applying Transformations We now present the overall algorithm for applying these transformations to a given JavaScript AST. The input to the algorithm is a list of transformation rules and the AST of the original JavaScript le. The algorithm consists of three steps: (a) Identifying AST nodes where transformations should be applied, i.e., translation sites; (b) rewriting the AST by modifying the subtrees rooted at the translation sites; and (c) adding code to the AST root to instantiate the generated WebAssembly modules. The algorithm outputs the transformed AST corresponding to the obfuscated JavaScript code. a) Identifying AST Nodes as Translation Sites: To identify translation sites, we perform a pre-order traversal of the AST starting at the root node. For each visited node n , the algorithm iterates through the list of transformation rules and checks which rules are applicable. A transformation rule $(L; t; p)$ is applicable if the node n is in the set L of code locations and if the precondition p holds for n . A set of translation site nodes is produced for each transformation rule. b) Rewriting AST Subtrees: After identifying all translation site nodes, the next step is to rewrite the subtrees rooted at these nodes. The algorithm applies transformations based on the size of the syntactic structures they target. Specifically, we iterate through the transformation rules in this order, applying each rule to all applicable subtrees before moving on to the next rule: T1-StringLiteral, T2-ArrayInitialization, T3-FunctionName, T4-CallExpression, T5-IfStatement, T6-ForStatement, T7-WhileStatement. This ordering ensures that transformations targeting finer-grained syntactic structures, such as string and array literals, are performed prior to transformations targeting coarser-grained structures, such as loops. If more coarse-grained transformation were applied first, the change could prevent more fine-grained transformations from being applied. For each rule $(L; t; p)$, the algorithm visits all translation site nodes and applies the transformation function t , which modifies the AST in-place and yields a WebAssembly module used in the rewritten code. The output of this step is the rewritten AST and a set W of WebAssembly modules. c) Adding WebAssembly Instantiation Code: The final step is adding code to instantiate the WebAssembly modules W . To this end, the algorithm inserts statements at the beginning of the script, i.e., at the root of the AST. For modules that are

instantiated synchronously, we encode each module in W as a base64 string and add statements that decode and instantiate the modules. For asynchronously instantiated modules, we serialize the modules to separate les and issue corresponding fetch requests in the code. For asynchronous translations, the algorithm additionally adds the async wrapper (described in Section IV-A) around the root of the AST to support asynchronous keywords in the remainder of the code. V. I

IMPLEMENTATION We implement Wobfuscator with Node.js (v14.17.2) and TypeScript. The tool relies on the Esprima (v4.0.1) [3] and Espree (v7.3.1) [2] packages to parse the JavaScript les and on Escodegen (2.0.0) [1] to convert the transformed AST back into a JavaScript le. The Wabt.js (1.0.23) [8] package is used to generate the WebAssembly modules used in the obfuscation. The data from our evaluation is available at <https://github.com/js2wasm-obfuscator/translator>. We make the implementation of Wobfuscator available upon request. We believe this strategy minimizes the threat of nefarious usage while also aiding researchers in independently reproducing our results, in conrming the identified weaknesses of existing malware detectors, and in serving as a foundation for future research on improving malware detectors. VI. E VALUATION We evaluate Wobfuscator and its ability to obfuscate mali- cious JavaScript using opportunistic translation to WebAssem- bly along the following main research questions:

RQ1 – Effectiveness : How effective is the approach at evading state-of-the-art JavaScript malware detectors and which transformations are most effective? How does our approach compare with other state-of-the-art obfuscators?

RQ2 – Correctness : Do our code transformations preserve the semantics of the transformed code?

RQ3 – Efcency : How much runtime and code size overhead do the transformations impose, and how long does applying the transformations take? To investigate these questions, we perform a comprehensive analysis on the effectiveness of our transformations in evading detection. We evaluate state-of-the-art detection tools against Wobfuscator on a large dataset of malicious and benign les. We evaluate the obfuscation advantage produced by Wobfuscator by comparing our approach against state-of-the- art open-source obfuscation tools. Lastly, we use the extensive test suites of widely used and mature npm modules to verify the correctness of our tool and demonstrate the runtime and code size overhead are acceptable for real-world usage. A.

Experimental Setup

1. Datasets:

Due to different requirements, we use dif- ferent datasets of JavaScript programs for different research questions. To answer RQ1, we need to train and apply state-of- the-art JavaScript malware detectors to large sets of real-world benign and malicious JavaScript code. Table II summarizes the datasets we use. The benign code consists of 149,677 les from the JS150k dataset [49]. The malicious code consists of 43,499 samples, with 2,674 samples from VirusTotal [7], 39,450 samples from the Hynek Petrak JavaScript malware collection [47], and 1,375 samples from the GeeksOnSecurity malicious JavaScript dataset [11]. These datasets are broken down further by the malware categories that they contain, such as trojans, ransomware, droppers. We list the breakdown of the malicious datasets in the rst four columns of Table IX. 8

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(34)*+,-./0--1+2/3145+./61784*99

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TABLE III RECALL OF MALWARE DETECTORS ON CODE OBFUSCATED BY W OBFUSCATOR . LOWEST RECALL IN BOLD .Technique Cujo Zozzle JaSt JStap (NGrams) JStap (Values)Baseline : No transformation 0.98 (5,548/5,649) 0.66 (3,598/5,453) 0.99 (5,076/5,108) 0.99 (4,483/4,524) 0.98 (4,439/4,524)Individual transformations:Sync, T1-StringLiteral 0.61 (1,623/2,644) 0.62 (3,387/5,453) 0.66 (3,393/5,108) 0.36 (1,539/4,257) 0.43 (1,839/4,257) Sync, T2-ArrayInitialization 0.94 (4,050/4,292) 0.66 (3,593/5,450) 0.85 (4,360/5,105) 0.86 (3,890/4,505) 0.89 (4,009/4,505) Sync, T3-FunctionName 0.67 (2,780/4,159) 0.65 (3,550/5,453) 0.69 (3,512/5,108) 0.57 (2,747/4,810) 0.72 (3,463/4,810) Sync, T4-CallExpression(a) 0.71 (3,040/4,285) 0.64 (3,507/5,453) 0.38 (1,943/5,108) 0.37 (1,723/4,633) 0.78 (3,613/4,633) Sync, T4-CallExpression(b) 0.58 (2,385/4,115) 0.63 (3,424/5,453) 0.44 (2,253/5,108) 0.23 (1,058/4,586) 0.73 (3,369/4,586) Sync, T5-IfStatement 0.82 (3,513/4,301) 0.64 (3,505/5,453) 0.89 (4,535/5,108) 0.83 (3,717/4,501) 0.93 (4,178/4,501) Sync, T6-ForStatement 0.90 (3,877/4,299) 0.66 (3,578/5,453) 0.92 (4,720/5,108) 0.87 (3,872/4,465) 0.98 (4,360/4,465) Sync, T7-WhileStatement 0.90 (3,904/4,321) 0.66 (3,598/5,453) 0.96 (4,882/5,108) 0.98 (4,410/4,502) 0.98 (4,412/4,502)Combined transformations:All sync (using T4(a)) 0.18 (416/2,255) 0.63 (3,450/5,450) 0.22 (1,104/5,105) 0.00 (1/4,235) 0.18 (766/4,235) All sync (using T4(b)) 0.19 (415/2,205) 0.63 (3,428/5,450) 0.18 (931/5,105) 0.00 (0/4,243) 0.08 (350/4,243) All async 0.28 (1,490/5,297) 0.65 (3,524/5,453) 0.20 (1,085/5,453) 0.00 (4/4,612) 0.22 (959/4,267) Fig. 4. Baseline recall vs. obfuscated recall. lexical analysis on malware les, so it learns the suspicious features of strings that are indicative of malware. Since T1- StringLiteral removes these identifiers from the le, Cujo is not able to find suspicious tokens in the le. Zozzle identifies malicious combinations of syntax features and the context they are used in. Since T1-StringLiteral removes the usage of malicious strings, such as shell code and dynamic code, malicious les that have the majority of their code leveraging these strings will have indicative features removed from their AST, allowing them to evade Zozzle. JaSt uses n-grams of AST syntax nodes (with a length of four by default) to detect malicious patterns. T4-CallExpression(a) replaces a single CallExpression node with multiple nodes handling the WebAssembly module instantiation. Since this translation adds more nodes than the length of sliding window, it can throw off the detection of known malicious n-gram patterns. JStap in its n-grams mode generates a program dependency graph (PDG) by adding edges representing data flow to the AST of an input le. Since T4-CallExpression(b) (and other transformations) route data flow through the generated WebAssembly module, it hinders the n-gram features that JStap can extract when performing its data flow analysis. JStap in its values mode is most affected by the T1-StringLiteral transformation since, in this mode, it relies on literals when selecting features to extract. T1-StringLiteral removes some literals. The lower part of Table III shows results from applying all transformation rules at once. We report results for three transformation combinations. “All sync (using T4(a))” and “All sync (using T4(b))” mean all synchronous transformations with T4(a) being used and with T4(b) being used, respectively. “All async” means all transformations in Table I (aside from T4(b)) in their asynchronous variant. We find that combining all transformation rules greatly reduces the recall of the detectors. In particular, with the “All sync (using T4(a))” set of transformations, Cujo, Zozzle, JaSt, JStap (NGrams), and JStap (Values) have a recall of 0.18, 0.63, 0.22, 0.00, and 0.18, respectively. Because of its performance and compatibility with the WebAssembly MVP language, we select “All sync (using T4(a))” to be the

default configuration for Wobfuscator. Figure 4 visualizes the results for “All sync (using T4(a))”. The main goal of our work is to reduce the recall of detectors, but we also measure their precision. The precision values for the transformations are listed in Table VII. For most applied transformations, the precision remains between 0.9 and 1.0. However, certain transformations can greatly impact the precision on some detectors, e.g., “All sync (using T4(a))” reduces JStap (N-Grams) precision to 0.5. This shows that while reducing the precision is not its main objective, Wobfuscator can reduce the precision of certain detectors.

1. Comparison with Other Obfuscators:

To demonstrate how Wobfuscator compares against currently available JavaScript obfuscators, we evaluate four obfuscation tools on the same dataset used in Section VI-B1. We collect the precision and recall values obtained by the various malware detection tools when evaluated on a dataset obfuscated by each tool. Similar to Section VI-B1, some detectors fail to parse certain obfuscated files, leading to different denominators in the values within the same detector column. Results: Table IV shows the recall values of the detection tools (columns) when run on code obfuscated by each of the four obfuscation tools described in Section VI-A3 (rows). The last row shows the best recall values obtained by Wobfuscator. The results show that Wobfuscator outperforms current obfuscators when compared on the recall reduction of malware detectors. The only exception occurs when Jfogs is evaluated against JaSt. In this case, Jfogs' recall rate of 0.00 outperforms Wobfuscator's recall rate of 0.18. Jfogs' obfuscation primarily replaces identifiers and literals with new intermediate variables, so Wobfuscator could be used to complement Jfogs. For example, Jfogs moves string literals into variables, but 10

--- page 13 ---

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--- page 14 ---

TABLE IV RECALL OF MALWARE DETECTORS ON CODE OBFUSCATED BY W OBFUSCATOR AND OTHER OBFUSCATORS .Obfuscator Cujo Zozzle JaSt JStap (N-Grams) JStap (Values)JavaScript Obfuscator 1.00 (4,406/4,415) 0.70 (3,807/5,453) 0.81 (4,153/5,108) 0.43 (2,005/4,717) 0.62 (2,947/4,717) Gnirts 0.98 (5,548/5,649) 0.66 (3,598/5,453) 0.99 (5,076/5,108) 0.99 (4,483/4,524) 0.98 (4,439/4,524) Jfogs 0.77 (3,515/4,562) 0.66 (3,584/5,453) 0.00 (26/5,453) 0.00 (16/5,025) 0.56 (2,826/5,025) JSObfu 1.00 (4,994/5,008) 0.84 (4,467/5,324) 0.29 (1,456/4,979) 0.01 (20/3,667) 0.66 (2,420/3,667) Wobfuscator (best recall) 0.18 (416/2,255) 0.62 (3,387/5,453) 0.18 (931/5,105) 0.00 (4/4,612) 0.08 (350/4,243)it does not alter or remove the strings from the file. Using the T1-StringLiteral transformation, the string literals can be completely removed from the JavaScript file, reducing the syntactic information available to the detectors.

1. Breakdown of Results by Malware Type:

The malicious datasets contain several different categories of malware, including cryptominers, trojans, and droppers. We provide a breakdown on the malware categories contained within our dataset. VirusTotal provides the malware type reported by the AV scanners. We reduce the number of malware categories presented by merging similar groups together, e.g., merging JS:Trojan.Gnaeus and JS:Trojan.Agent into a Trojan category. For GeeksOnSecurity, we use the directory names to identify which samples are exploits kits and which are JavaScript droppers. The

Hynek Petrak dataset does not provide metadata on the samples, so we scan the les with ClamAV to obtain the malware categories. To demonstrate the effectiveness of Wobfuscator in obfuscating a diverse set of malware samples, we measure the reduction in the recall of malware detectors for different malware categories in the dataset. For space reasons, we list only the minimum recall rates observed within the malware categories among all of the transformations. Results: Columns 5-9 of Table IX present the recall values obtained by the malware detectors (columns) when samples from each malware category (rows) are obfuscated. In addition to the recall rate, each cell lists the number of les correctly marked as malicious by the detector over the number of malicious les the detector tested within that category. Columns marked with '-' signify that the detectors are unable to parse any of the test les within the malware category. For the Phishing malware category, no samples appear in our test dataset, so no recall values are available. The results on the ve largest malware groups (Downloader, Misc., Trojan, Malware, and Exploit) show that Wobfuscator can significantly reduce recall rates across diverse malware categories.

C. Correctness of the Transformations (RQ2) The transformations we apply change the syntactic structure of the program. Naturally, such changes could affect program semantics, potentially making the obfuscated program behave differently from the original program and thus breaking functional correctness. This gives rise to two questions, which are in tension with each other: First, do we preserve functional correctness of the input program, i.e., are our code transformations semantics-preserving? Second, how often are the transformations applied? To validate that the correctness of the program is preserved, we leverage the comprehensive test suites of existing widely used JavaScript projects. We apply the transformations to the tested code and then validate if the transformed code still passes its tests. Addressing the second question, a trivial solution to correctness would be to transform only a very small set of code locations, preserving semantics at the expense of obfuscating less code. Thus, we also evaluate how often each transformation rule is applied. The validation setup differs between the synchronous and asynchronous variants of the transformations. In the synchronous case, we can simply apply the transformations to any existing code. To validate the correctness of the asynchronous transformations, the projects must be modied to support asynchronous execution as described in Section IV-C. Since automatically turning arbitrary JavaScript code into asynchronous code is non-trivial, we instead focus on an NPM project, node-fetch , that is already asynchronous, so applying the asynchronous transformations is simplified. We use this project to validate the asynchronous variants and use the other ve projects to validate the synchronous variants. Results: The results for the test suite runs are shown in Table V. This table lists the tested project, its version, and the number of translation sites where rules T1–T7 are applied to. The last two columns list the total number of tests in the test suite, and the number of tests that are impacted by at least one transformation. All tests in each project pass successfully, showing that our obfuscations are semantics-preserving. Columns 4-11 of Table V show the number of transformed code locations that meet the preconditions of the transformation out of the total number of available code locations relevant to the transformation, regardless of whether they satisfy the preconditions. The last two columns of Table V show that of the 2,017 unit tests in the ve test suites, 1,844 of them (91.42%) rely on a function that is impacted by at least one transformation rule. The results show that our transformation rules are applicable to code locations used in the real-world.

D. Efficiency in Terms of Runtime and Code Size (RQ3) The Wobfuscator transformations we propose reimplement native JavaScript functionalities in WebAssembly modules, such as calling a function,

performing a while loop, initializing an array, etc. As a result, there will be an impact on the performance of the translated programs. To quantify the performance impact of our transformations, we use the test suites of the six modules described in Section VI-C. In addition, the code size increase caused by the transformations is also analyzed, counting both added JavaScript and WebAssembly code.

1. Translation Runtime:

First, we measure the time taken for performing the transformations on the project les. The times are measured with the time command available in 11

--- page 15 ---

TABLE V C ORRECTNESS VALIDATION RESULTS . Project Version LoC T1 T2 T3 T4(a) T4(b) T5 T6 T7															
Total # of Tests # of Tests ImpactedValidation of synchronous transformations:Lodash 5.0.0															
21,178	193/801	160/208	0/0	87/467	299/467	47/187	0/0	38/61	408	322	Chalk 4.1.0	319	64/68	0/23	
2/2	21/108	26/108	4/25	1/1	2/2	54	54	Commander 7.2.0	1,153	155/163	0/28	0/0	130/394	87/394	
91/153	4/4	2/3	632	592	Debug 4.3.2	505	141/149	2/8	0/0	20/95	43/95	16/29	2/5	0/0	14
13	Async 3.2.0	787	30/57	0/21	0/0	89/209	108/209	31/86	5/6	5/8	675	659	Validation of asynchronous transformations:Node-Fetch 3.0.0beta.10		
970	174/212	1/17	0/0	49/264	-	26/94	0/0	0/0	234	204	Total -	4,152	757/1,450	163/305	2/2
396/1,537	654/1,537	189/480	12/16	47/74	2,017	1,844									

TABLE VI EFFICIENCY OF TRANSFORMATIONS . Project															
Translation time	Execution time	Code size (bytes)	LoC	Time	Original	Overhead	Original	Overhead	Synchronous	Validation	Lodash	21,178			
29.58s	3.51s	+25.81%	135,402	+139.84%	Chalk	319	0.81s	4.03s	+7.01%	14,935	+166.70%				
Commander	1,153	0.51s	3.97s	+49.95%	74,269	+146.96%	Debug	505	1.14s	0.61s	+3.24%	21,395			
+154.70%	Async	787	5.42s	16.83s	+2079.21%	28,925	+363.52%	Asynchronous	Validation	Node-Fetch	970	16.03s	4.11s	+14.76%	51,960
127.16%	Average	4,152	8.92s	5.68s	+31.07%	54,481	+170.42%								

Linux, averaged over ten repetitions. We compute the total transformation time of a project by summing the times to convert each JavaScript file used in the project. The transformation time results are presented in Table VI. The table shows that for the largest project, Lodash with 21,178 lines of code, the average time to apply all of the synchronous transformations is 29.58 seconds. For the smallest project, Chalk with 319 lines of code, the average time to apply the transformations is only 0.81 seconds. In addition, we find that among all the projects in Table VI, the average time to apply all of the transformations is only 8.92 seconds. These low transformation times show that Wobfuscator is practical for JavaScript obfuscation.

1. Execution Time Overhead:

The execution overhead time is the increase in runtime to complete the execution of the test suites of the transformed projects. We use the time command to measure the runtime of the project test suite before and after the transformations are applied, reporting averages over ten repeated measurements. The execution time results are presented in Table VI. Our transformations add a performance overhead that ranges from an increase of 3.24% to an increase of 2,079%. While the highest overhead number is large, it is important to note that this large runtime originates from one test within the async project that concurrently applies an asynchronous function to a collection of 1,048,576 elements. In most cases, it is unlikely that malware samples will follow such an execution pattern that incurs this large overhead. On average, Wobfuscator adds a performance overhead of 31.07%.

1. Code Size Overhead:

The code size overhead is the increase in code size between the original le and transformed output among all code les within a project. Table VI lists the percentage of growth in code size compared with the original size. On average, applying all of the transformations among the project, the code size increased by 170.42%. Overall, the code size overhead is acceptable for practical applications.

VII. D ISCUSSION This section discusses the limitations and possible mitigations to defend against WebAssembly-based obfuscation.

A. Limitations Wobfuscator targets malware detectors based on static analysis, and despite its effectiveness in bypassing them, is unlikely to be equally effective for dynamic analysis-based detectors. The transformations move some behavior into Web-Assembly while leaving the ultimate runtime behavior intact. That is, a dynamic detector that, e.g., observes browser API calls made by a website will observe the same behavior with and without our obfuscation. However, in practice static detectors are much easier to deploy (e.g., as network proxies or browser extensions), whereas observing dynamic behavior is more complex to set up and expensive at runtime. Another limitation is that the approach applies transformations only to some of the given code. If a code location does not fulll the preconditions for a specic transformation, then it cannot be transformed. Conservatively guarding transformations is crucial to ensure that our approach preserves the semantics of the given code, but also limits its applicability. Finally, our obfuscation relies on WebAssembly being avail-able in the browser. With WebAssembly support in 94% of all installed browsers, this limitation is likely to be acceptable in practice. To ensure that the obfuscated malware runs as expected, an attacker could check for WebAssembly support and load the obfuscated code only if the language is supported.

B. Mitigations We discuss three mitigation strategies aimed at detecting malware despite our obfuscation. The rst is dynamic analysis-based malware detection. Because our approach preserves the original JavaScript behavior, many runtime characteristics that dynamic detectors focus on [53] are not affected by the obfuscation. WebAssembly code invokes web APIs through JavaScript, which means the call will be visible to any runtime analysis that wraps the API functions or intercepts them within the browser. However, dynamic malware detectors often im-pose a non-negligible runtime overhead and may miss malware that hides its malicious behavior in specic congurations. 12

--- page 16 ---

The second mitigation strategy is based on the defender knowing the details of our obfuscation. Since the WebAssem-bly usage in the obfuscated code, e.g., loading many small modules, may be abnormal, it is possible to dene heuristic rules to detect that Wobfuscator was applied. In a similar vein, one could include code obfuscated with our technique in the training data used to learn a malware classier. The main drawback of these mitigations is that obfuscation does not imply maliciousness. There are legitimate reasons for obfuscating code, e.g., protecting intellectual property. Hence, classifying all code obfuscated by our technique, or any other obfuscation technique, as malicious is likely to cause an unacceptably high number of false positives. Finally, the third mitigation strategy is to jointly analyze JavaScript and WebAssembly. For detectors based on tradi-tional program analysis, both static or dynamic, a joint analysis would reason about how data and control ows between the two languages. Likewise, learning-based detectors, such as those used in our evaluation, could feed code in both languages into their models. We are not aware of any existing malware detector with support for WebAssembly but hope that our work will

raise awareness that a joint analysis would be useful.

VIII. RELATED WORK

Obfuscation Studies and Techniques:

Obfuscation techniques have been observed in various programming languages and software domains for both malicious and benign purposes. Previous works categorize obfuscation techniques applied on malicious code [69] while others compare the effectiveness of different obfuscation techniques [22], [32], [61]. Some studies have analyzed the usage of obfuscation techniques specifically in JavaScript code by investigating the obfuscation techniques used in real-world malicious and benign files [66], [55]. There is little work proposing new obfuscation attacks for JavaScript code. Fass et al. [26] construct HideNoSeek which rewrites the ASTs of malicious programs into the AST of known benign programs to avoid detection. The authors evaluate HideNoSeek on 91,020 samples against VirusTotal, Yara, JaSt, Zozzle and Cujo. The obfuscation technique is able to achieve a 99.98% false negative rate against the detectors.

Malware Detection:

An active area in academic research produces static analysis techniques designed to identify malicious behavior even in the presence of obfuscated code. One class of static detection tools use lexical and syntactic information derived from JavaScript files in order to identify features that indicate malicious code [50], [24], [16], [20], [54]. Techniques build on this syntactic information by incorporating control-flow and data-flow analysis [27] or by adding dynamic analysis to confirm the presence of malware [67], [59]. Other static detection techniques analyze the JavaScript source code through machine-learning and deep-learning approaches [63], [46]. Wobfuscator can impact the detection rates of these detectors since it reduces the syntactic information available. In addition, some behavior is moved to WebAssembly modules, which are ignored by these detectors. Other malware detection techniques dynamically analyze programs to identify malicious behaviors. Some techniques focus on collecting runtime statistics to construct models that identify malware [23], [53], [68]. Other techniques leverage symbolic execution [39] or forced execution [38] to trigger malware hidden behind complex input sequences. Wobfuscator is unlikely to reduce the detection rate against these detectors as we do not significantly change the runtime behavior.

Obfuscation Detection:

Some existing work only focuses on detecting obfuscation rather than obfuscated malware. NOFUS [35] and JSOD [13] use syntactic and contextual information as features for a machine learning classifier to detect obfuscation. Sarker et al. [53] develop a hybrid approach by instrumenting browser APIs and determining whether the traced API call corresponds to a static code location.

WebAssembly and WebAssembly Security:

Haas et al. [31] explain the motivation and benefits of introducing a new byte code language to the Web. Several works investigate security aspects of WebAssembly, e.g., unsolicited cryptomining in the browser [40], [45], [52] and how to detect and defend against it [37], [51], [62]. Lehmann et al. show that source-level memory vulnerabilities may propagate to WebAssembly binaries [41], a problem that affects many real-world binaries [33]. However, neither of those works use WebAssembly to hide arbitrary JavaScript behavior from inspection. Future joint malicious code detectors for JavaScript and WebAssembly could build upon the Wasabi framework [42] or taint tracking frameworks for WebAssembly [29], [57].

IX. CONCLUSION

Much work has focused on identifying JavaScript malware using static analysis. However, these techniques ignore recent web standards available to attackers, namely WebAssembly. To bypass static detectors, we present Wobfuscator, an obfuscation approach built on a set of seven transformation rules that opportunistically translate specific parts of JavaScript code into functionally identical WebAssembly modules. We evaluate our transformations against four state-of-the-art static JavaScript malware detectors and show that our approach effectively reduces the recall on real malware samples. We

show that our technique outperforms other obfuscation tools only based on JavaScript. Finally, we use the test suites of six NPM packages to validate the correctness of our transformations and show their low performance overhead. Our results show that current static detectors are ineffective against techniques that implement cross-language code obfuscation, motivating future work on addressing this challenge.

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--- page 17 ---

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ForStatement 0.93 (3,877/4,161) 0.97 (3,578/3,674) 1.00 (4,720/4,724) 1.00 (3,872/3,873) 1.00 (4,360/4,361) Sync, T7-WhileStatement 0.93 (3,904/4,188) 0.97 (3,598/3,694) 1.00 (4,882/4,886) 1.00 (4,410/4,411) 1.00 (4,412/4,413) Combined transformations: All sync (using T4(a)) 0.59 (416/700) 0.97 (3,450/3,546) 1.00 (1,104/1,108) 0.50 (1/2) 1.00 (766/767) All sync (using T4(b)) 0.59 (415/699) 0.97 (3,428/3,524) 1.00 (931/935) 0.00 (0/1) 1.00 (350/351) All async 0.84 (1,490/1,774) 0.97 (3,524/3,620) 1.00 (1,085/1,089) 0.80 (4/5) 1.00 (959/960)

TABLE VIII P RECISION OF MALWARE DETECTORS ON CODE OBFUSCATED BY OTHER OBFUSCATORS . Obfuscator Cujo Zozzle JaSt JStap (NGrams) JStap (Values) javascript-obfuscator 0.94 (4,406/4,690) 0.98 (3,807/3,903) 1.00 (4,153/4,157) 1.00 (2,005/2,006) 1.00 (2,947/2,948) Gnirts 0.95 (5,548/5,832) 0.97 (3,598/3,694) 1.00 (5,076/5,080) 1.00 (4,483/4,484) 1.00 (4,439/4,440) jfogs 0.93 (3,515/3,799) 0.97 (3,584/3,680) 0.87 (26/30) 0.94 (16/17) 1.00 (2,826/2,827) JSObfu 0.95 (4,994/5,278) 0.98 (4,467/4,563) 1.00 (1,456/1,460) 0.95 (20/21) 1.00 (2,420/2,421)

TABLE IX M ALWARE CATEGORY BREAKDOWN OF DATASETS AND DETECTOR RECALL RATES . Malware Type Dataset Breakdown Malware Detector Recall VirusTotal GeeksOnSecurity Hynek Petrak Cujo Zozzle JaSt JStap (N-grams) JStap (Values) Downloader 14 0 216 190.10 (102/988) 0.67 (1772/2629) 0.11 (279/2629) 0.00 (0/2440) 0.00 (2/1787) Misc 167 21 135 270.35 (257/731) 0.64 (1100/1732) 0.36 (625/1732) 0.00 (0/1082) 0.06 (60/1082) Trojan 1618 0 205 80.00 (0/201) 0.61 (305/499) 0.00 (0/499) 0.00 (0/453) 0.06 (25/453) Malware 0 0 154 60.00 (0/109) 0.29 (66/231) 0.22 (51/231) 0.00 (0/180) 0.02 (3/125) Exploit 6 1029 00.00 (0/151) 0.00 (0/151) 0.00 (0/151) 0.00 (0/151) 0.00 (0/151) Ransomware 12 0 7000.00 (0/56) 0.05 (3/59) 0.00 (0/3) 0.00 (0/56) - Cryptominer 665 0 01.00 (1/1) 0.99 (68/69) 0.00 (0/69) 0.00 (0/1) 0.00 (0/1) Dropper 0 323 00.00 (0/17) 0.50 (26/45) 0.11 (5/45) 0.00 (0/45) 0.00 (0/45) Hijacker 112 0 00.00 (0/12) 0.94 (17/18) 0.00 (0/18) 0.00 (0/12) 0.00 (0/11) Riskware 21 0 01.00 (1/1) 1.00 (5/5) 0.00 (0/5) - - Redirector 15 0 00.00 (0/3) 0.25 (1/4) 0.00 (0/4) 0.00 (0/3) 0.00 (0/3) Clicker 25 0 0- 1.00 (4/4) 0.00 (0/4) - - Iframe 24 0 0- 1.00 (4/4) 0.00 (0/4) 1.00 (1/1) 1.00 (1/1) Clickjack 13 0 0- 1.00 (1/1) 0.00 (0/1) - - Phishing 3 0 0- - - - 16